

# Family 1.0.tf: land and coast at ten kilometres and five days a dataset design note — the raw observations of the land surface, its coasts and its imagery

blauewelt/earth — design note

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## Abstract

Family 1.0.tf is the land-and-coast family of fine-granularity observation stores: measurements at **ten kilometres or finer and five days or finer** over the continents and a coastal band, on family 10's registry, tiers and token. It is designed for what a model of the land surface has to predict — where forest will be cleared, how much carbon a canopy holds and how fast that changes, where water will stand, what burns, what is planted, where cities grow — as well as for the forcing the four ocean goals need from the land that borders the sea. Three properties of the land record set the design. The fine record is mostly *points* (38,870 hourly and 132,437 daily weather stations, 2,932 radiosonde sites, 6,474 tide-gauge records, coastal buoys and radars, gliders, flux towers, half a billion fire detections, and the LiDAR footprints that measure vegetation in three dimensions), and they fit tier P with one widening of the time column. The daily *fields* at  $0.05^\circ$  are 20–30% valid and need the sharded, tiled, compressed layout family 1.gf introduces. And the *imagery* — optical at 10–30m across forty years and four instruments on six satellites, radar at C and L band in dual and quad polarisation, sub-metre orthophotos where a country publishes them — is 0.01–3 petabytes a year per sensor and can never be pixels: it enters as a scene catalogue now and as cached local-codec tokens later, per E-078's tier T. Every candidate was checked against its producer on 2026-09-16; a licence rule sorts each store onto a public track (the public Hub repository, readable by anyone) or a private track (built and held for this project's own use, never republished), so a source's terms decide where it lives and not whether it is used; and the note gives the stores, the arithmetic, a three-phase ladder (phase A  $\approx 0.7$  TB), the falsifiers, and the decisions left to Chris.

## 1 Purpose

The programme builds a forward model of the Earth's surface state: encode what is observed at a place into a compact vector, predict it forward, read out anything (Chris, 2026-08-19). Families 7.2 and 10 gave that model the ocean at  $0.25^\circ$  and the ocean's point measurements; family 1.gf gives it the global fine scale. Nothing at ten kilometres or finer over land exists in any family yet, and the land is where the finest and most consequential predictions are asked for. This family exists for three uses, in this order.

**Land change as a prediction target.** Deforestation and forest degradation (where the next clearing happens, at 10–30 m, weeks ahead); the carbon a canopy holds and how it changes (biomass and canopy height, and the rate at which a forest can plausibly regrow after loss); surface water and floods; burned area; crop type and condition; urban growth. Each has an observed read-out today — alert systems at 10–30 m, annual loss maps back to 1990, LiDAR biomass, water-extent maps — and each is a fine-scale, land-only quantity that no ocean family can supply.

**The land as forcing for the ocean goals.** The atmosphere over the continents that border the Atlantic and the Pacific was fitted by the reanalyses the model has seen so far; the stations, balloons, snow and land-temperature fields below are what those reanalyses were fitted to, and a model that has only seen the fit cannot learn where the fit fails (family 1.gf's argument for ship reports, on land).

**Coastal sea level and currents as read-outs of the AMOC and El Niño.** Tide gauges since 1846 along the US east coast see the Florida Current; Pacific island gauges see Kelvin waves weeks before Niño-3.4 moves; high-frequency radar measures the surface current at 6km hourly where the boundary currents meet the shelf.

## 2 The rule, the two domains, and the two tracks

**The rule** is family 1.gf’s: an observation with spatial support  $\leq 10$  km (a point counts) and temporal support  $\leq 5$  days; a derived product only where it carries raw observations otherwise inaccessible; coarser cadence only where critical, and named as an exception. Read-out *targets* are held to the spatial rule and not the temporal one — an annual loss map is what the model is asked to predict, not what it is shown.

**The two tracks.** A source’s licence decides where its store lives, not whether it is built. A store whose terms allow redistribution (`licence.redistribution` of **yes** or **attribution**) goes on the *public track*: the public Hugging Face dataset repository, readable by anyone and by every agent that works on this project. A store whose terms forbid it (**no**) goes on the *private track*: the same nine arrays and the same registry entry, held in a private repository that only this project’s own jobs can read, used for training and evaluation here as a single licensee and never republished, forwarded or mirrored. The registry field `distribution` (**public** or **private**) records the track, and family 1.gf §4 gives the mechanism: where the private repository is, how a rented box reads it for one job, and what may be published from a model trained on it. Until 2026-09-16 the design had a licence *gate* that refused such sources outright, because every store was assumed to be republished; Chris removed the assumption the same day (“we can manage to build private datasets, and train on them without re-distributing”), and the five sources the gate had refused are re-sorted here. Two are set aside for reasons the rule itself gives: **HadISD** (Non-Commercial Government Licence) is a quality-controlled derivative of the hourly reports `ghcnh` carries raw under CC0, and **MSWEP** (CC BY-NC) is a merged product whose gauges are reachable through `ghcnd`. Three become private-track stores: **ISMN** (3,300 soil-moisture stations — “re-export or transfer of the original data to a third party is prohibited”) as `ismn`; **GRDC** (10,000 river gauges, no internet redistribution) as `river_grdc` beside the public `river`, whose CC BY 4.0 Caravan subset of 5,356 catchments stays public; and **ECA&D** (25,919 European stations, non-commercial, no redistribution) as `ecad`, phase C and only for the series GHCN-Daily lacks. The same sorting applies inside a store: `tide`’s research-only GESLA sources (CMEMS, CV, UZ) and the South African gauges are no longer dropped by a filter but written by the same build to a private sibling, `tide.private`; and Planet-NICFT’s non-commercial tropical basemaps join `cat_submetre` on the private track once their new-phase terms are read. One source stays out, on access rather than licence: **Dynamic World**, Google’s 10 m per-scene land-cover classification (CC BY 4.0), exists only inside Earth Engine, and an export at Sentinel-2’s scene volume would be a catalogue of a derived product.

**Two domains.** *Terrestrial*: the continents, inland water and ice sheets. *Coastal*: a band of  $\pm 100$  km around the GSHHG shoreline (the Global Self-consistent Hierarchical High-resolution Geography set, public domain), rasterised once at  $0.05^\circ$  into a static `dist_coast` channel. For a gridded group “coastal” means the tiles whose minimum distance to coast is under 100 km; a point store holds every row it has and the consumer bounds by distance. The band is a build parameter recorded in the registry, not a property of the data.

**What is held already and not repeated.** Family 7.2’s land channels at  $1^\circ$  and its statics; family 10.1’s tropical moorings and along-track sea level; family 10.2’s daily  $0.1^\circ$  fishing effort (E-081), densest on the shelves.

## 3 Time axis, footprint and token

Time is family 10’s five-day bin from 1982-01-01, with the exact second in `time_s`; every row or pixel becomes the E-078 token — value, offset from the anchor in kilometres and days, and the two footprint fields  $\log_2 f_p = \log_2(\text{support km}/27.83)$  and  $\log_2 f_t = \log_2(\text{support days}/5)$ . Family 1.gf’s conventions apply unchanged:  $-4$  is the *label* for a point instrument with no areal support, a real support carries its true value even below  $-4$  (a 30 m pixel is  $-9.9$ , a 25 m LiDAR footprint  $-10.1$ , an hour  $-6.9$ ), and a coarse value is one token at its own centre, never copied into finer cells. Two land-specific points. A LiDAR footprint’s *vertical* axis is not a footprint field but a channel axis (§4.2). And an annual target map (a loss year, a biomass map) carries  $\log_2 f_t = \log_2(365/5) = +6.2$  and is filed under the bin of its period’s end, so that a target is never read as an input from inside its own year.

## 4 The stores

Checked 2026-09-16 against producers' pages, the NASA Common Metadata Repository, and live S3 and directory listings; counts marked *measured* were summed from the producer's inventory files; a dagger (†) marks an estimate. Bytes: a tier-P row costs  $27 + 2C$  (schema 2) or  $31 + 2C$  (schema 3, int64 `time_s`, for records starting before 1914); a sharded-grid pixel costs about  $v \times 2C \times 1.2$  with  $v$  the valid fraction.

## 4.1 Point networks, tier P

| store        | source; channels ( $C$ )                                                                                                                                                                                                                                          | points                                                 | cadence                      | record; rows                                                                                   | stored                       | licence                                                    | role; phase                                            |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------|--------------------------------------------------------|
| ghcnd        | GHCN-Daily v3.34 (NOAA NCEI): TMAX, TMIN, PRCP, SNOW, SNWD ( $C = 5$ , schema 3)                                                                                                                                                                                  | 132,437 stations, 42,984 active (measured)             | daily                        | 1763 $\rightarrow$ ; $\approx 1.0 \times 10^9$ station-days (3.2 B element rows)               | $\approx 40$ GB              | CC0                                                        | forcing; A                                             |
| igra         | IGRA v2.2 radiosondes (NCEI): T, dew-point depression, $u$ , $v$ , height at the 16 mandatory levels ( $C = 80$ , schema 3)                                                                                                                                       | 2,932 stations, 807 active (measured)                  | 00 / 12 UTC                  | 1905 $\rightarrow$ ; 52.3 M soundings                                                          | 10 GB                        | US government                                              | forcing, El Niño; A                                    |
| tide         | GESLA-4 (4.1 download, July 2026) via the UHSLC ERDDAP, 36 sources incl. UHSLC Fast Delivery and Research Quality, NOAA CO-OPS: relative sea level ( $C = 1$ , schema 3)                                                                                          | 6,474 records, 127,623 station-years                   | $\leq$ hourly                | 1800s $\rightarrow$ 2026; $1.2\text{--}1.5 \times 10^9$ on the public track                    | 40–50 GB                     | mixed; the research-only sources go to <b>tide_private</b> | AMOC, El Niño; A                                       |
| tide_private | the GESLA-4 sources with research-only terms (CMEMS, CV, UZ) and, with SANHO’s permission, the South African gauges; same schema as <b>tide</b>                                                                                                                   | the station-years the public sort removes <sup>†</sup> | $\leq$ hourly                | as <b>tide</b>                                                                                 | 10–20 GB <sup>†</sup>        | research-only: private track                               | AMOC, El Niño; A (same build)                          |
| ndbc         | NDBC buoys and C-MAN stations (NOAA): wind $u$ , $v$ , gust, wave height, peak and mean period, direction, pressure, air T, SST ( $C = 10$ )                                                                                                                      | 1,345 stations, 829 with 2025 data (measured)          | hourly, 10-min recently      | 1970 $\rightarrow$ 2025; $3\text{--}5 \times 10^8$ <sup>†</sup>                                | 15–25 GB                     | US government                                              | SST, forcing; A (TAO sites excluded: in <b>gtmba</b> ) |
| gliders      | IOOS Glider DAC (2,540 datasets) + OceanGliders GDAC (1,120 deployments): T, S, O <sub>2</sub> , chlorophyll at the 16 Roemmich–Gilson pressures ( $C = 64$ )                                                                                                     | $\approx 5 \times 10^6$ profiles <sup>†</sup>          | profiles every $\approx 3$ h | 2003 $\rightarrow$                                                                             | 0.8 GB                       | open / CC BY 4.0                                           | AMOC, SST, carbon; A                                   |
| flux         | FLUXNET Shuttle (2026-04-30): eddy-covariance towers from AmeriFlux, ICOS, TERN, ChinaFLUX: NEE, GPP, Reco, LE, H, Rn, Ta, VPD, SWC, P ( $C = 10$ )                                                                                                               | 700+ sites, $\approx 6,000$ site-years                 | half-hourly                  | 1991 $\rightarrow$ ; $\approx 1.05 \times 10^8$                                                | 5 GB                         | CC BY 4.0 (six Tier-2 and Legacy sites excluded)           | land carbon read-out; A                                |
| fire         | FIRMS active fire (NASA): MODIS C6.1 1 km (2000-11 $\rightarrow$ ), VIIRS S-NPP / NOAA-20 / NOAA-21 375 m (2012 $\rightarrow$ ): brightness, FRP, confidence, day/night ( $C = 4$ )                                                                               | detections                                             | per overpass                 | MODIS $1.1 \times 10^8$ , VIIRS $\approx 4 \times 10^8$ (2023 measured: 4.66 M / 21.75 M)      | 20 GB                        | NASA open <sup>†</sup>                                     | read-out, carbon; A                                    |
| river        | USGS NWIS daily discharge + GRDC-Caravan ( $C = 1$ , schema 3)                                                                                                                                                                                                    | 24,976 US sites, 8,691 active (measured); 5,356        | daily                        | 1800s $\rightarrow$ ; $\approx 4 \times 10^8$                                                  | 14 GB                        | public domain / CC BY 4.0                                  | read-out, freshwater; B                                |
| river_grdc   | GRDC daily discharge, the full archive ( $C = 1$ , schema 3)                                                                                                                                                                                                      | > 10,000 gauges, 160 countries <sup>†</sup>            | daily                        | 1800s $\rightarrow$ ; $\approx 1.5 \times 10^8$ <sup>†</sup>                                   | $\approx 5$ GB <sup>†</sup>  | GRDC terms: private track                                  | read-out, freshwater; B                                |
| ismn         | ISMN (International Soil Moisture Network, TU Wien): soil moisture and soil temperature in up to five depth slots per station, the station’s own depths kept in <b>platforms.json</b> , NaN where it has no sensor, with the network’s quality flags ( $C = 10$ ) | 3,300 stations, 80+ networks                           | hourly                       | 1952 $\rightarrow$ ; $\approx 3 \times 10^8$ <sup>†</sup>                                      | $\approx 15$ GB <sup>†</sup> | ISMN terms: private track                                  | forcing; the in-situ truth for soil moisture; B        |
| ecad         | ECA&D (KNMI) daily non-blended series — TX, TN, TG, RR, SS, SD, HU, PP ( $C = 8$ , schema 3) — for stations absent from GHCN-Daily only                                                                                                                           | subset of 25,919 series <sup>†</sup>                   | daily                        | 1800s $\rightarrow$                                                                            | $\leq 20$ GB <sup>†</sup>    | non-commercial, no redistribution: private track           | forcing; C                                             |
| ghcnh        | GHCN-hourly v1 (NCEI), successor of the frozen ISD: T, dew point, pressure, $u$ , $v$ , precipitation, snow depth, cloud ( $C = 8$ , schema 3)                                                                                                                    | 38,870 stations, 10,313 active (measured)              | hourly / synoptic            | 1700s $\rightarrow$ ; $7.60 \times 10^9$ reports (measured)                                    | 360 GB                       | CC0                                                        | forcing; B                                             |
| hfr          | HFRNet US (6 / 2 / 1 km) + European HFR Node (22 systems): $u$ , $v$ at the cell centre ( $C = 2$ )                                                                                                                                                               | $\approx 12.5$ k valid 6 km cells per hour, US         | hourly                       | 2012 $\rightarrow$ US, $\approx 2009 \rightarrow$ EU; $3.5\text{--}5 \times 10^9$ <sup>†</sup> | 110–155 GB                   | free / CC BY 4.0                                           | coastal currents, AMOC; B                              |
| alticap      | ALTICAP coastal altimetry, Jason-3 20 Hz, 0–500 km from coast (AVISO/ODATIS): anomaly, quality ( $C = 2$ )                                                                                                                                                        | along-track, $\approx 350$ m                           | $\approx 10$ -day repeat     | 2016-02 $\rightarrow$ 2021-07 only                                                             | 3 GB                         | CC BY 4.0                                                  | AMOC; B                                                |

Table 1: The point networks. Station stores ship a twelfth file, **platforms.json**, with each hash’s source identifier, elevation and the source’s station metadata (§6).

## 4.2 LiDAR footprints: the vertical structure of vegetation, tier P

Optical and radar imagery see a canopy from above; only LiDAR measures it in three dimensions, and the two spaceborne instruments do so as sparse footprints along orbital tracks — exactly the case a point store was built for. The vertical axis goes into  $C$  the way pressure levels do for Argo and radiosondes: the footprint’s own relative-height profile, not a single “height”.

| store   | source; channels ( $C$ )                                                                                                                                                                             | footprint                                                                            | record; rows                                                                                                                                                              | stored                                                             | licence     | role; phase                            |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------|----------------------------------------|
| gedi    | GEDI on the ISS (LP DAAC L2A/L2B version 3, ORNL L4A v2.1): ground elevation, RH25, RH50, RH75, RH90, RH98, canopy cover, plant-area index, footprint biomass and its error, quality ( $C = 11$ )    | 25 m footprints every 60 m along 8 tracks, $\pm 51.6^\circ$ ; $\log_2 f_p = -10.1$   | 2019-04 $\rightarrow$ 2023-03 and 2024-04 $\rightarrow$ 2025-07 released (stored on the ISS between); “more than 10 billion high-quality observations”; L2A 106 TB native | 4–10 $\times 10^9$ quality rows $\times$ 49 B $\approx$ 0.2–0.5 TB | EOSDIS open | canopy structure, biomass; B           |
| icesat2 | ICESat-2 ATLAS (NSIDC ATL08 release 7): terrain height, canopy height, canopy cover, photon counts per 100 m segment ( $C = 6$ ); ATL07/ATL10 sea-ice height and freeboard in the polar coastal band | $\approx 11$ m footprints at 0.7 m spacing, 6 beams, $\pm 88^\circ$ ; 100 m segments | 2018-10 $\rightarrow$ ; ATL08 4.4–4.9 TB/yr native; segments $\approx 4 \times 10^9$ /yr over land <sup>†</sup>                                                           | $\approx 2.8 \times 10^{10} \times 39$ B $\approx 1.1$ TB          | NASA open   | structure beyond $51.6^\circ$ , ice; C |

Table 2: The two orbital LiDARs. GEDI’s version 3 (2026-07-01) improved geolocation to about 10 m; the instrument’s status after July 2025 is not stated by the mission and is treated as ongoing.

## 4.3 Daily fields at $0.05^\circ$ , tier G sharded

| store         | source; channels ( $C$ )                                                                                                                                                                                                                                             | grid                              | record; native                                                               | stored                                                                                   | licence   | role; phase                                                                                 |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------|
| 1st05         | MOD11C1 v061 (Terra) land-surface temperature day and night with QC ( $C = 4$ , $F = 5$ ); MYD11C1 (Aqua) and VJ121C1 / VJ221C1 (NOAA-20 2018 $\rightarrow$ , NOAA-21 2023 $\rightarrow$ ) as continuation groups                                                    | $0.05^\circ$ , $7200 \times 3600$ | Terra 2000-02-27 $\rightarrow$ ; $\approx 45$ MB/day, 9,591 granules         | 25.9 M px $\times$ 9,700 d, $v \approx 0.29 \times 0.5$ : $\approx 350$ GB per satellite | NASA open | forcing, the land twin of SST; A (Terra), B (Aqua, VIIRS $\approx 350 + 160$ GB) forcing; A |
| snow05        | MOD10C1 v61 (NSIDC): snow-cover fraction, cloud fraction as uint8 percent ( $C = 2$ ); VNP10C1 / VJ110C1 continuation                                                                                                                                                | $0.05^\circ$                      | 2000-02-24 $\rightarrow$ ; 4–6 MB/day                                        | $\approx 60$ GB (Terra; classes compress far below the float formula)                    | cite      |                                                                                             |
| ref105        | MOD09CMG v061 daily surface reflectance bands 1–7 and state QA ( $C = 8$ ); VJ109CMG / VJ209CMG continuation                                                                                                                                                         | $0.05^\circ$                      | 2000-02-24 $\rightarrow$ ; $\approx 600$ MB/day, 5.5 TB per satellite native | $v \approx 0.15$ : $\approx 700$ GB (Terra), +330 GB VIIRS                               | NASA open | vegetation, snow, burn state; albedo; B                                                     |
| ims1k         | NOAA IMS daily Northern-Hemisphere snow and ice chart (NSIDC G02156), 1 km class (uint8, $C = 1$ ); 4 km from 2004, 24 km from 1997                                                                                                                                  | $24024^2$ polar stereographic     | 1 km 2014-12-02 $\rightarrow$ ; 2.8 MB/day gz                                | classes compress $\approx 50 \times$ : $\approx 50$ GB                                   | cite      | forcing; B                                                                                  |
| chirps05 (E4) | CHIRPS v3.0 pentad precipitation (UCSB CHC): infrared + climatology + gauges, land $60^\circ\text{S}$ – $60^\circ\text{N}$ ( $C = 1$ ); its six-per-month pentads carried as a <code>frame.table</code> , each frame filed under the family bin holding its midpoint | $0.05^\circ$ , $17.3$ M px        | 1981 $\rightarrow$ ; $\approx 3,300$ pentads                                 | $v \approx 0.3$ : $\approx 40$ GB                                                        | CC BY 4.0 | forcing, teleconnection read-out; B                                                         |

Table 3: The daily fields. Terra and Aqua are still producing (end late 2026 / 2027, date not fixed); Suomi-NPP data end 2026-11-01, so the VIIRS continuation is NOAA-20 and NOAA-21; MODIS Collection 7 replaces v061 when published.

## 4.4 The image catalogue, tier T

Every source below is 0.01–3 petabytes a year; none is stored as pixels. Each is a tier-P *catalogue* store: one row per scene-date with `bin`, `time_s` (acquisition), `lat`, `lon` (footprint centre), `platform` (the hashed scene identifier), `qc` (the producer’s processing baseline), `fp` (the tile’s footprint; a 110 km Sentinel-2 tile is  $+2.0$ ), and  $C = 5$ : cloud fraction, valid fraction, sun zenith or radar incidence, area, and a **sensor code** — which instrument on which platform, from a code table in `store.json`. A sidecar `assets.parquet` maps the hash to the STAC identifier and asset URLs, so the token builder of E-078c can fetch exactly the scenes a cell needs. The token cache itself (E-078 §4:  $k$  tokens of 64 dimensions per tile-date from a local codec conditioned on the sensor code and its geometry) is a later tier-P store whose size E-078c measures.

| store             | source                                                                                                                                                                                                                                                                                                                                                                                     | scenes                                                                      | record; native volume                           | licence                                                     | note                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------|
| cat_landsat       | Landsat Collection 2 Level 2, <b>all sensors</b> : Landsat 4 and 5 TM (1982 / 1984 → 2012), 7 ETM+ (1999 → 2024), 8 and 9 OLI/TIRS (2013 / 2021 →); 30 m reflectance, 60–120 m thermal (USGS)                                                                                                                                                                                              | ≈ 1,400–1,500 per day today <sup>†</sup> ; ≈ 10 <sup>7</sup> in the archive | 1982 →; 0.3–0.5 PB/yr today <sup>†</sup>        | public domain                                               | the forty-year record, one sensor code per row                                  |
| cat_hls           | HLS v2.0 S30 + L30, 30 m harmonised Sentinel-2 and Landsat 8/9 (LP DAAC)                                                                                                                                                                                                                                                                                                                   | 2.71 M + 1.75 M granules in 2025 (measured)                                 | 2013-04 / 2015-11 →; ≈ 0.7 PB/yr                | NASA open                                                   | the harmonised sample of the same record                                        |
| cat_s2            | Sentinel-2 L2A, 10 / 20 / 60 m; S2B + S2C (S2A retired 2026-03) (ESA / CDSE)                                                                                                                                                                                                                                                                                                               | 12.7–15.5 k per day (measured), ≈ 600 MB each                               | 2015-07 →; ≈ 3 PB/yr                            | Copernicus open                                             | CDSE account; 12 TB per 30 days                                                 |
| cat_s1            | Sentinel-1 IW GRDH dual-pol VV/VH (HH/HV at high latitude), 10 m pixels; S1C + S1D (S1A retired 2026-06-30); the SLC is catalogued beside it for the dual-pol covariance                                                                                                                                                                                                                   | 982 per day (measured), 1.72 GB                                             | 2014-10 →; GRD ≈ 620 TB/yr, SLC ≈ 1.6 PB/yr     | Copernicus open                                             | C-band polarimetry: dual, never quad                                            |
| cat_nisar         | NISAR L2 GCOV provisional (2026-07-20 →): L-band geocoded covariance, quad-, dual- and compact-pol modes, 10 / 20 / 80 m posting (ASF)                                                                                                                                                                                                                                                     | 28,456 granules in August 2026                                              | 2026-06-17 →; 0.7–1 PB/yr                       | NASA open <sup>†</sup>                                      | the only free quad-pol L-band                                                   |
| cat_palsar (E5)   | JAXA PALSAR / PALSAR-2 global 25 m mosaics v2.6 (HH, HV; 1° tiles) and the ScanSAR L2.2 scenes                                                                                                                                                                                                                                                                                             | ≈ 15 k tiles per year                                                       | mosaics 2007–2010, 2015–2025; ScanSAR 2014-08 → | JAXA research terms, ©JAXA, registration                    | annual: an exception, the L-band record before NISAR                            |
| cat_olci          | Sentinel-3 OLCI L2 WFR 300 m, coastal band; GlobColour OLCI 300 m L3 (Copernicus 009.103) as the gridded form, which phase C may also store as a sharded tier-G group over the coastal band (oc300.coast, chlorophyll and flags, ≈ 1.6 TB)                                                                                                                                                 | 457 per day, 150 GB/day (measured)                                          | 2016-04 →; ≈ 55 TB/yr                           | Copernicus open                                             | coastal tokens                                                                  |
| cat_viirs         | VIIRS I-band L1B 375 m (VNP02IMG / VJ102IMG, LAADS)                                                                                                                                                                                                                                                                                                                                        | 225–242 granules/day                                                        | 2012 / 2018 →; ≈ 15 TB/yr per satellite         | NASA open                                                   |                                                                                 |
| cat_ecostress     | ECOSTRESS L2T LSTE 70 m thermal from the ISS, irregular                                                                                                                                                                                                                                                                                                                                    | 1.37 M granules in 2024                                                     | 2018-07 →; ≈ 10 TB/yr                           | NASA open                                                   | no cadence; catalogued only                                                     |
| cat_submetre (E6) | sub-metre and metre-scale orthophotos where a country publishes them openly: USDA NAIP 0.6–1 m (US, 2–3-yearly, public domain), swisstopo SWISSIMAGE 10 cm (Switzerland, 3-yearly, open government data); Planet-NICFI 4.77 m monthly tropical basemaps (non-commercial; programme contract ended 2025-01, basemaps in Earth Engine to 2026-08, new-phase terms unverified; private track) | regional                                                                    | 2009 / 2017 / 2015 →                            | public domain / OGD / non-commercial (NICFI: private track) | regional, multi-year: an exception; no free sub-metre source is global or daily |

Table 4: The catalogued imagery. The ten catalogues together are  $\approx 1.6 \times 10^8$  rows and  $\approx 7$  GB; pixels are never copied.

## 4.5 Read-out targets

Targets are what the footprint-queried decoder (E-078 §5) is asked to predict at the target’s own footprint. They are held to the spatial rule only.

| store              | source; channels                                                                                                                                                                                                                                                           | resolution | record                               | stored                          | licence                                 | phase |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------|---------------------------------|-----------------------------------------|-------|
| <b>alerts</b>      | forest-disturbance alerts as point rows (date, position, confidence, system): GLAD-L (Landsat 30 m, tropics, 2015 →), GLAD-S2 (10 m, 2019 →), RADD (Sentinel-1 10 m, 6–12 day, humid tropics, WUR, CC BY 4.0), OPERA DIST-ALERT (30 m global, 2023 →)                      | 10–30 m    | 2015 →; rows $\approx 10^{9\dagger}$ | $\approx 30$ GB                 | CC BY 4.0 (GLAD terms unverified)       | A     |
| <b>lossyear</b>    | Hansen Global Forest Change v1.13 tree cover 2000 and loss year 2001–2025 (30 m, CC BY 4.0); JRC Tropical Moist Forest v2025 annual change 1990–2025 (30 m, free without restriction); annual uint8 sharded grids                                                          | 30 m       | 1990 / 2001 → 2025                   | loss is sparse: $\approx 30$ GB | CC BY 4.0 / free                        | A     |
| <b>burned500</b>   | MCD64A1 v6.1 burned-area date (500 m, monthly, LP DAAC)                                                                                                                                                                                                                    | 500 m      | 2000–11 →                            | $\approx 10$ GB                 | unrestricted                            | A     |
| <b>cat_gfm</b>     | Copernicus Global Flood Monitoring: Sentinel-1 flood extent per pass at 20 m, archive to 2015, $\approx 2$ M scenes; OPERA DSWx-HLS / DSWx-S1 30 m water extent                                                                                                            | 20–30 m    | 2015 / 2016 →                        | catalogue, < 1 GB               | free (GFM terms unverified) / NASA open | A     |
| <b>biomass100</b>  | ESA CCI Biomass v7.0 (2026-05-21): above-ground biomass and its standard deviation, 100 m, years 2005–2012 and 2015–2024; GEDI L4B 1 km biomass 2019–2023                                                                                                                  | 100 m      | annual                               | 760 GB native; $\approx 0.4$ TB | ESA CCI terms                           | C     |
| <b>landcover10</b> | annual 10 m land cover as an annual target: Esri / Impact Observatory 2017–2025 (CC BY 4.0), ESA WorldCover 2020 and 2021 (CC BY 4.0), MapBiomass Brazil 1985–2024 (30 m, CC BY 4.0); crops: USDA CDL 30 m annual (public domain), EUCROPMAP 2018 / 2022, WorldCereal 2021 | 10–30 m    | annual                               | $\approx 0.2$ TB <sup>†</sup>   | CC BY 4.0                               | C     |

Table 5: The read-out targets. GHSL built-up (100m, five-yearly, 10m for 2018) and WSF Evolution (30m annual 1985–2015, CC BY 4.0) are the urban targets and join **landcover10** in phase C.

## 4.6 Statics

**static.fine**: GEBCO\_2026 elevation and bathymetry at 15" (7.5 GB as float16, public domain); WorldCover 2021 class fractions at 0.05° (11 channels, 0.6 GB, CC BY 4.0); **dist\_coast** at 0.05° from GSHHG; and, for the 10m tile tier, Copernicus DEM GLO-30 referenced on AWS rather than copied. About 8 GB, phase A.

## 5 Missing values and quality

A channel not measured is NaN; a value outside its physical bound is NaN and counted, never clipped; the source’s own quality flag is kept in **qc** and nothing is homogenised. Fields carry their quality layers as channels (**1st05**’s QC day and night; **ref105**’s state QA) rather than as a mask, so the consumer decides the threshold. Absence is information: a station that stopped, a LiDAR track that was not flown, a cloudy day — all read as  $n_R = 0$  or a NaN pixel, and the model is told so.

## 6 Design decisions

**Licence decides where a store lives, not whether it exists.** The first draft of this note refused four of the best land networks because every store was to be republished, and accepted having no soil-moisture store as the price. The two-track rule of §2 removes the price: a source that may not be redistributed is built once, held privately, used here as a single licensee, and never leaves. The cost that remains is named in family 1.gf §4 — a private store is invisible to the other agents and to anyone reproducing the work — so a result that depends on one says so, and the public track stays the default wherever a licence allows it.

**Vertical structure is stored natively, as the LiDAR’s own profile.** A canopy is a three-dimensional object and imagery cannot see into it. **gedi** therefore keeps the footprint’s relative-height profile (RH25 to RH98), cover and plant-area index as channels, and **icesat2** its terrain and canopy heights per segment — the vertical axis in *C*, as depth is for Argo. A single “canopy height” derived product would discard exactly what the instrument was flown for.

**State and event are both stored, so that a rate is measurable.** A forest can be cleared in a day and takes decades to regrow; biomass and height change slowly and within physical limits. The family keeps the *state* (height and biomass from LiDAR, annual biomass and land-cover maps) and the *event* (dated alerts, loss years, burned-area dates) as separate stores with their own timestamps, so that a model’s predicted change can be checked against what growth and loss can physically do in the interval. Whether such bounds are enforced in the model or only measured is the model’s decision; the data design makes the measurement possible.

**One catalogue across forty years of sensors, never blended.** `cat_landsat` carries Landsat 4, 5, 7, 8 and 9 from 1982 with a sensor code per row, and `cat_hls` carries NASA’s harmonised sample of the recent part; the two coexist. The catalogue does not blend or calibrate across sensors — that is the local codec’s job, conditioned on the sensor code and geometry (E-078 §4), and the falsifier in §9 asks whether it learned the shift. Radar is stored the same way: C-band dual-pol from Sentinel-1 (there is no quad-pol mode), L-band dual-pol from PALSAR and dual- and quad-pol from NISAR, each under its own code.

**The catalogue before the codec.** Ten catalogues at 7 GB turn the tile tier’s scene counts into measured numbers before a single token is computed, and are the only way to size E-078c against reality; they are phase A even though the tokens wait.

**The coastal band is a build parameter.**  $\pm 100$  km covers the shelf break almost everywhere and the radars’ 200 km reach; it is recorded in the registry so a rebuild at another width is a rebuild, not a redesign.

**Station elevation lives in the platform table.** Hashing a station identifier would lose the one number that makes a 2 m temperature comparable with a  $0.05^\circ$  skin temperature; `platforms.json` keeps it, with the source’s own metadata, beside the nine arrays.

**Annual products are exceptions, and named.** E4 CHIRPS (pentad, derived, gauge-bearing, CC BY 4.0) is the one derived field admitted as an input; E5 PALSAR mosaics (annual, the L-band record before NISAR) and E6 regional sub-metre orthophotos (multi-year cycles) are admitted to the catalogue only. Targets are outside the temporal rule by construction.

**What the imagery is for.** The catalogue lists optical at 10–30 m across six sensors, radar in two polarisations at two bands, LiDAR footprints and sub-metre orthophotos where they exist — the raw inputs a land representation must be learned from. Family 0.9.tf is the alternative that takes a published embedding — computed from the same optical and C-band imagery and trained against the same LiDAR, L-band and climate sources — in place of the pixels; the two families are meant to be compared, not merged.

## 7 Validation

Every array’s sha256 against `store.json`;  $N$  from the file sizes; `bin` monotone and equal to `floor_divide(times, 432000)`; the CSR index reproduced; for sharded grids, every index entry decompressing to a tile of the declared shape and NaN fraction, and per-year frame counts equal to the producer’s listing; for catalogues, scene counts per day against the producer’s own catalogue; for `tide`, the station-years sorted to `tide.private` reported. A day absent upstream is a zero-length frame with a counter, never a silent gap; a fetch lane refuses an empty listing or a short download and records any relaxation in `degraded`.

## 8 Limitations

`hfr`’s archive size is unverified because the UCSD server refused connections on the day of checking. GEDI’s operating status after July 2025 is not stated by the mission. `icesat2` at native segments is a terabyte and is phase C; a thinned version would be derived. The sub-metre tier is regional and multi-year by the nature of what is published openly. The tile tokens depend on a codec that does not yet exist (E-078c); until it does, the tile tier is a catalogue. Dynamic World cannot be built for lack of a bulk copy. A private-track store can be checked by nobody outside the project, so §7’s validation is the only certificate it has, and a

published result that rests on one names it. Nothing here is dispatched on an unmeasured size: the probe of §9 replaces every estimate first.

## 9 Next steps: the ladder

| phase | stores                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | stored           | access                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | <code>ghcnd</code> , <code>igra</code> , <code>tide</code> and <code>tide_private</code> , <code>ndbc</code> , <code>gliders</code> , <code>flux</code> , <code>fire</code> , <code>1st05</code> (Terra), <code>snow05</code> (Terra), nine of the ten <code>cat_*</code> catalogues ( <code>cat_palsar</code> waits for phase B), <code>alerts</code> , <code>lossyear</code> , <code>burned500</code> , <code>cat_gfm</code> , <code>static_fine</code> , the registry | $\approx 0.7$ TB | seven need no account at all ( <code>ghcnd</code> , <code>igra</code> , <code>tide</code> , <code>ndbc</code> , <code>gliders</code> , <code>lossyear</code> , <code>static_fine</code> ); <code>flux</code> and <code>fire</code> ’s NOAA-20/21 archive go through a portal request; the NASA stores and catalogues need an Earthdata login on a hosted runner; the Sentinel catalogues a CDSE account; the Landsat one a USGS login or a requester-pays AWS read; <code>1st05</code> is assembled on one verified box (\$2–5); under \$10; <code>tide_private</code> is written by the <code>tide</code> build to the private repository |
| B     | <code>gedi</code> , <code>ghcnh</code> , <code>hfr</code> , <code>river</code> , <code>river_grdc</code> , <code>ismn</code> , <code>alticap</code> , <code>refl05</code> , <code>ims1k</code> , <code>chirps05</code> , <code>cat_palsar</code> , the Aqua and VIIRS continuations of <code>1st05</code> and <code>snow05</code>                                                                                                                                        | 2.3–2.7 TB       | \$20–30 in box time; the Hugging Face conversation about terabyte repositories first; <code>ismn</code> and <code>river_grdc</code> are registered downloads on a hosted runner, published to the private repository                                                                                                                                                                                                                                                                                                                                                                                                                       |
| C     | <code>icesat2</code> , <code>biomass100</code> , <code>landcover10</code> , <code>ecad</code> , the tile tokens over E-078c’s coastal cells, <code>oc300_coast</code> (the 300 m colour of <code>cat_olci</code> stored over the band, $\approx 1.6$ TB)                                                                                                                                                                                                                 | $\approx 3$ TB   | after E-078c’s verdict                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table 6: The build ladder. Phase A is the eight point stores ( $\approx 150$  GB, one of them private), the two Terra fields ( $\approx 410$  GB), the catalogues and targets ( $\approx 80$  GB) and the statics.

**The probe.** One month per source through the real adapter on a hosted runner: rows per day and distinct platforms; valid fraction and compressed bytes per tile; scene counts per day against the producer’s catalogue; throughput; and for `tide` the station-years the licence sort sends to `tide_private`.

**Two falsifiers.** For the *forcing and coastal* stores, family 1.gf’s test: at anchors where the store has coverage, its tokens in the cone must lower the held-out masked-token loss on the goal’s read-out against the same anchors without them, three seeds; parity means the reanalysis and the coarse tensor already carried it. For the *land targets* the test is reversed and is the family’s main one: the embedding trained on this family’s inputs must predict each target at its own footprint better than climatology and better than the same read-out from family 7.2 alone — next year’s alerts from this year’s state, biomass from imagery plus LiDAR, burned area from temperature and reflectance — and, for the forty-year record, the codec’s tokens of the same place from Landsat 5 and Landsat 8 must agree with each other more than with a different place, which is the test that the sensor shift was learned rather than memorised.

## 10 Decisions for Chris

1. **The name:** 1.0.tf as given.
2. **The two tracks:** `ismn`, `river_grdc`, `ecad`, `tide_private` and NICFI on a private track rather than refused, so the soil-moisture store returns. Pick: yes, in the private repository of family 1.gf §4.

3. **The coastal band:**  $\pm 100$  km. Pick: 100 km.
4. **GEDI in phase B at full footprint resolution** (0.2–0.5 TB) rather than thinned. Pick: full; thinning is derived.
5. **Landsat back to 1982 in the catalogue**, all sensors, HLS beside it. Pick: yes.
6. **Targets in phase A** (alerts, lossyear, burned500, cat\_gfm), cheap and the reason the family exists. Pick: yes.
7. **Exceptions E4–E6:** CHIRPS as input; PALSAR mosaics and regional orthophotos in the catalogue. Pick: admit all three; NICFI on the private track once its new-phase terms are read.
8. **Tide gauges:** GESLA-4 filtered per source, else UHSLC’s two sets. Pick: GESLA filtered.

## Terms used above, in one line each

**Tier G / P / T** family 10’s three storage tiers: gridded fields; points, profiles and tracks; tiles of imagery held as a catalogue plus codec tokens. **STAC** SpatioTemporal Asset Catalog, the standard in which producers list scenes. **CSR** compressed sparse row — an offsets array saying where each five-day bin’s rows start. **GSHHG** the Global Self-consistent Hierarchical High-resolution Geography shoreline set. **GHCNh / GHCN-Daily / ISD** NOAA’s hourly and daily station archives and the frozen hourly archive GHCNh replaces. **IGRA** the Integrated Global Radiosonde Archive. **GESLA / UHSLC / CO-OPS** the global hourly tide-gauge compilation, the University of Hawaii Sea Level Center, NOAA’s tide-gauge service; **CV, UZ, CMEMS** three GESLA source codes with research-only terms; **SANHO** the South African Navy Hydrographic Office. **NDBC / C-MAN** NOAA’s buoy centre and its coastal automated stations. **HFRNet / IOOS** the US high-frequency coastal radar network and the Integrated Ocean Observing System, which also runs the glider data-assembly centre (DAC). **FLUXNET / AmeriFlux / ICOS** networks of eddy-covariance towers measuring the exchange of CO<sub>2</sub>, water and heat between land and air; **NEE / GPP / Reco / LE / H / Rn / VPD / SWC** net ecosystem exchange, gross primary production, ecosystem respiration, latent and sensible heat, net radiation, vapour-pressure deficit, soil water content. **FIRMS / FRP** NASA’s fire-detection service and fire radiative power. **NWIS** the US Geological Survey’s water archive. **ALTICAP / ODATIS / AVISO** a coastal altimetry product and the French ocean-data services that host it. **GEDI / ICESat-2 / ATLAS / ATL08** NASA’s two spaceborne LiDARs, the ICESat-2 instrument, and its land-and-canopy product; **RH<sub>n</sub>** the relative height below which  $n$  % of a LiDAR footprint’s returned energy lies — RH98 is the canopy top. **CMG** MODIS’ Climate Modeling Grid, 0.05°. **IMS** the US National Ice Center’s daily snow and ice chart. **CHIRPS** the Climate Hazards infrared precipitation with stations. **L1B / L2 / L2A / L3 / GRD / SLC / GCOV** calibrated radiances; retrievals; Sentinel-2’s surface reflectance; gridded; Sentinel-1’s detected and complex radar products; NISAR’s geocoded covariance. **TM / ETM+ / OLI / TIRS** the Landsat 4–5, 7, and 8–9 instruments. **HLS** Harmonized Landsat Sentinel-2. **PALSAR / NISAR** JAXA’s and NASA-ISRO’s L-band radars. **OLCI WFR / VIIRS / ECOSTRESS** Sentinel-3’s ocean-colour product, the JPSS imager, and the ISS thermal imager. **NAIP / NICFI** the US aerial orthophoto programme and Norway’s tropical-forest satellite programme. **GLAD / RADD / DIST-ALERT** forest-disturbance alert systems from the University of Maryland, Wageningen, and NASA OPERA. **JRC TMF** the EU Joint Research Centre’s tropical moist forest maps. **MCD64A1** MODIS burned area. **GFM / DSWx** Copernicus’ flood monitoring and OPERA’s surface-water maps. **ESA CCI** ESA’s Climate Change Initiative. **GHSL / WSF** the Global Human Settlement Layer and the World Settlement Footprint. **CDL / EUCROPMAP / WorldCereal** crop-type maps from the USDA, the JRC and ESA. **ISMN / ECA&D / GRDC** the soil-moisture network, the European station set and the Global Runoff Data Centre — private-track stores; **HadISD / MSWEP** the Met Office hourly station set and a merged precipitation product, set aside as derivatives; **Caravan** GRDC’s CC BY subset, public. **Public / private track** a store anyone may read from the public repository, or one held for this project’s own use because its licence forbids redistribution (family 1.gf §4). **GEBCO** the General Bathymetric Chart of the Oceans. **E-078 / E-078c / E-081** the multi-granularity design, its first coastal tile-codec experiment, and the fishing-effort store. **AMOC / SST** the Atlantic Meridional Overturning Circulation and sea-surface temperature. **CC0 / CC BY / CC BY-NC** public-domain dedication, attribution, and non-commercial Creative Commons licences. **MODIS / VIIRS** the imaging radiometers on Terra and Aqua, and on Suomi-NPP, NOAA-20 and NOAA-21. **LP DAAC / NSIDC / EOSDIS** NASA’s land-processes and snow-and-ice data centres and the system that serves them. **CDSE** the Copernicus Data Space Ecosystem, ESA’s Sentinel archive. **ERDDAP** a data server that answers tabular queries over HTTP. **LSTE** land-surface temperature and emissivity. **GRDH** ground-range-detected high-resolution. **GLO-30** the Copernicus 30 m elevation model. **OGD** open government data. **TERN / ChinaFLUX** the Australian and Chinese flux-tower networks. **Dynamic World** Google’s 10 m per-scene land-cover classification, Earth Engine only. **NaN** not-a-number, the stored value for “not measured”.

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